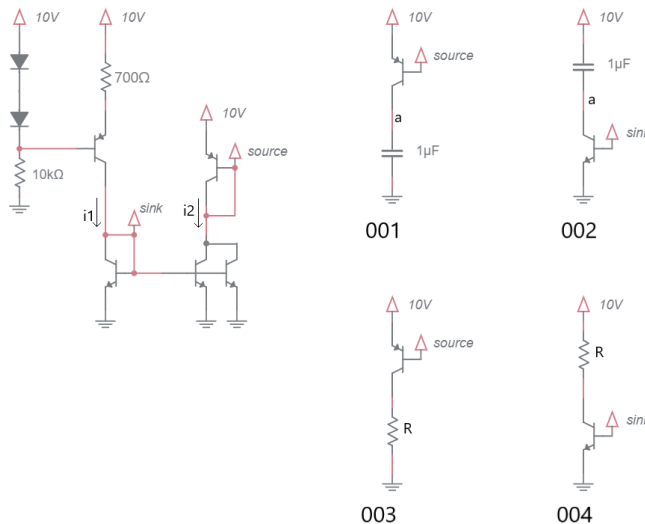




## BJT Current Sources and mirrors

If the BJT is in the active region, assume  $V_{be} = 0.7V$ ,  $\beta = 100$ ,  $i_c = i_e$ .



1. Determine the currents  $i_1$  and  $i_2$ .

|                               |                               |
|-------------------------------|-------------------------------|
| <b><math>i_1 = 1mA</math></b> | <b><math>i_2 = 2mA</math></b> |
|-------------------------------|-------------------------------|

2. What is the range (upper and lower) of resistance in the circuits 003 and 004 to keep their BJTs in the active region?

|            |                                   |                                           |           |                                 |
|------------|-----------------------------------|-------------------------------------------|-----------|---------------------------------|
| <b>003</b> | <b>saturation @ 9.7V with 2mA</b> | <b><math>R = V/I = 4.85k\Omega</math></b> | <b>to</b> | <b><math>R = 0\Omega</math></b> |
| <b>004</b> | <b>saturation @ 9.7V with 1mA</b> | <b><math>R = V/I = 9.7k\Omega</math></b>  | <b>to</b> | <b><math>R = 0\Omega</math></b> |

3. Plot the volage at point **a** in circuits 001 and 002 assuming the capacitor is discharge at time 0. Use the following questions to guide yourself to the answer.

- a. Using the voltage/current equation for a capacitor, find  $dv/dt$  for 001 and 002.

|            |                                                          |
|------------|----------------------------------------------------------|
| <b>001</b> | <b><math>dV/dt = i/C = 1mA/1\mu F = 10^3V/s</math></b>   |
| <b>002</b> | <b><math>dV/dt = i/C = 2mA/1\mu F = 2*10^3V/s</math></b> |

- b. Since the initial voltage drop across the capacitors is 0V, what is the initial value of **a**?

|            |                             |
|------------|-----------------------------|
| <b>001</b> | <b><math>a = 0V</math></b>  |
| <b>002</b> | <b><math>a = 10V</math></b> |

c. Since the final voltage drop across the capacitors is 10V, what is the final value of  $a$ ?

|            |                             |
|------------|-----------------------------|
| <b>001</b> | <b><math>a = 10V</math></b> |
| <b>002</b> | <b><math>a = 0V</math></b>  |

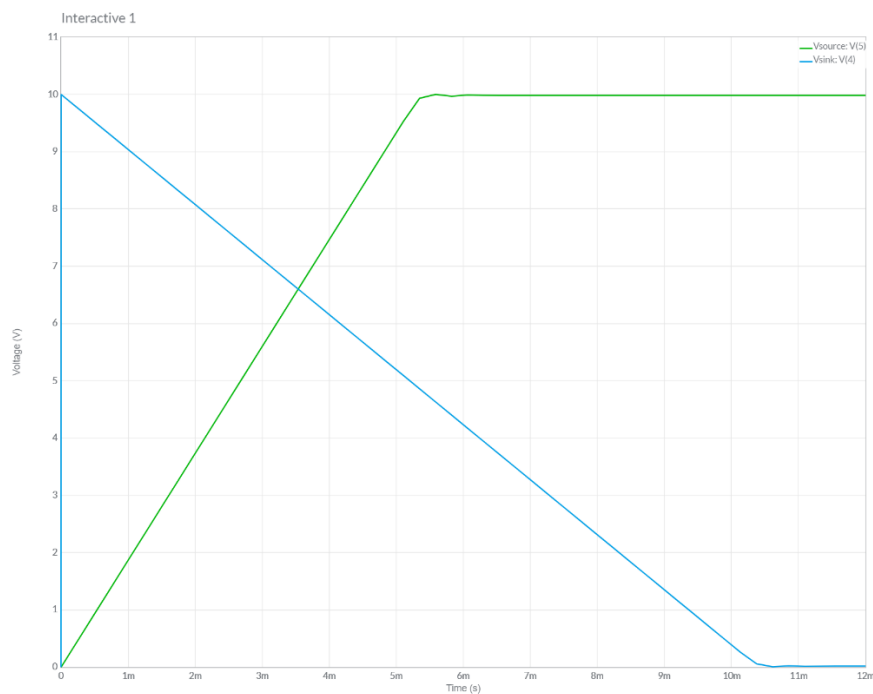
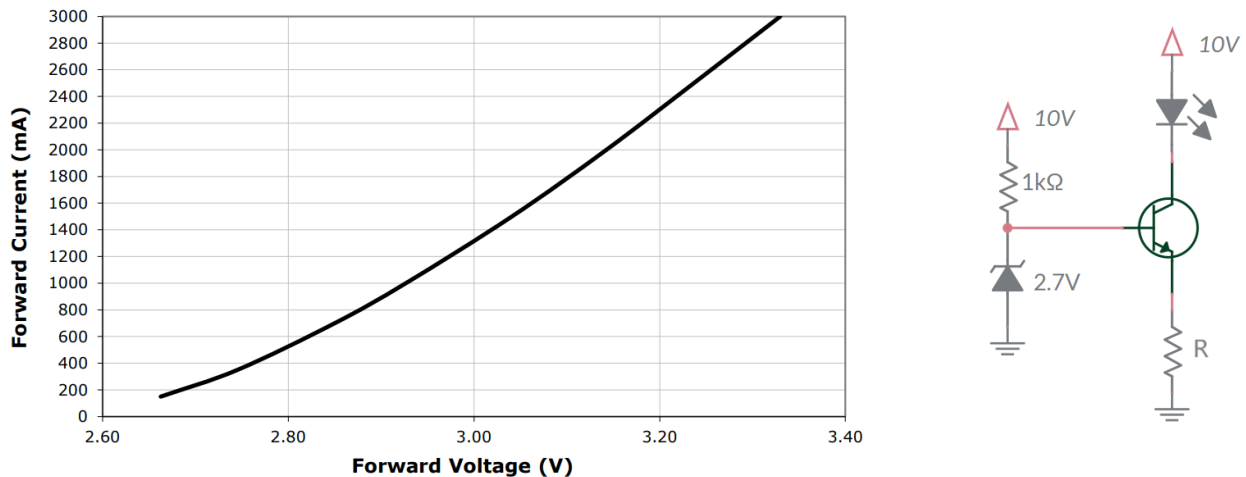


Figure 1: Plot the voltage at node  $a$  for the circuits shown in 001 and 002

## Practical Application – Power dissipated by BJT

You are required to illuminate a Cree Xlamp XM-L2 High Power LED using a current source. The LED has a I vs. V curve is shown at left in Figure 2. Your current source, shown at right in Figure 2, uses a TIP31 (a high powered NPN transistor) and a 2.7V Zener diode. You need to run the LED at 2 amps.



**Figure 2: (Left) I/V curve for a Cree Xlamp XM-L2 High Power LED. (Right) The constant current source used to drive the LED.**

- a. What is the emitter voltage?

$$2.7V - 0.7V = 2.0V$$

- b. What value of emitter resistor, R, will produce the needed 2A through the LED?

$$V = IR \quad 2V = 2A R \quad R = 1\Omega$$

- c. What is the voltage drop across the LED at 2A?

**According to Figure 3 (Left) the voltage drop across the LED is 3.15V**

- d. What is the power dissipated by the LED at 2A?

$$P = IV \text{ so } 3.15V * 2A = 6.3W$$

- e. What is the power dissipated by the resistor R at 2A?

$$P = IV = 2A * 2V = 4W$$

- f. What is the voltage across the BJT at 2A?

$$10V - 3.15V - 2V = 4.85V$$

- g. What is the power dissipated by the BJT at 2A?

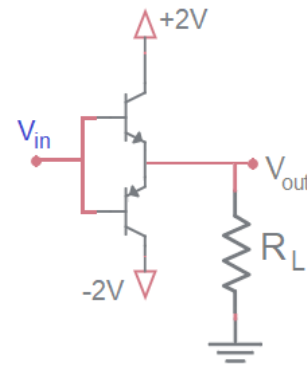
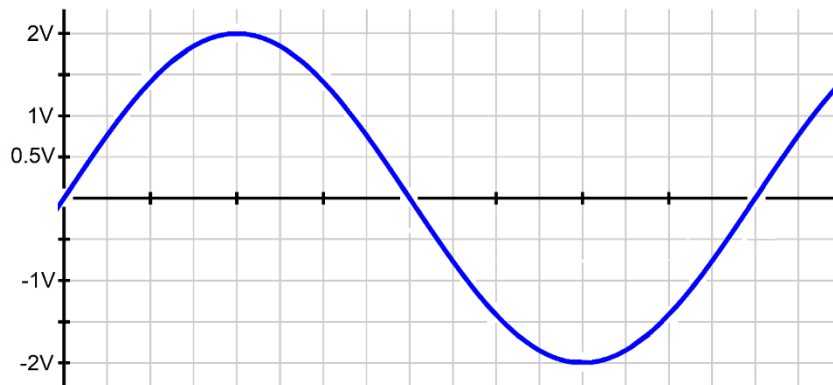
$$P = IV = 2A * 4.85V = 9.7W$$

- h. How can you reduce the power dissipation of the BJT while keeping 2A running through the LED? Do not change the circuit topology.

**Reduce the voltage supply down from 10V but no lower than the 3.15V for the LED.**

### Practical Application - Cross Over Distortion

The blue sine wave shown in the graph below is applied as the input,  $V_{in}$ , to the circuit shown below at right. Determine the output,  $V_{out}$ , and graph it on the graph below. The value of the load resistor,  $R_L$  is unimportant.



One method to determine the solution is to apply specific values of the sine wave to the circuit and determine how the circuit behaves. Let's start:

Let  $V_{in} = 2.0V$  In this case there is a path from  $V_{in}$  through the upper BJT's base emitter junction, through the  $R_L$  to ground. Thus,  $V_{out}$  is one  $V_{BE}$  (diode) drop below 2V or  $2V - 0.7V = 1.4V$ . Note the lower BJT has its base emitter junction reverse biased and thus, is off.

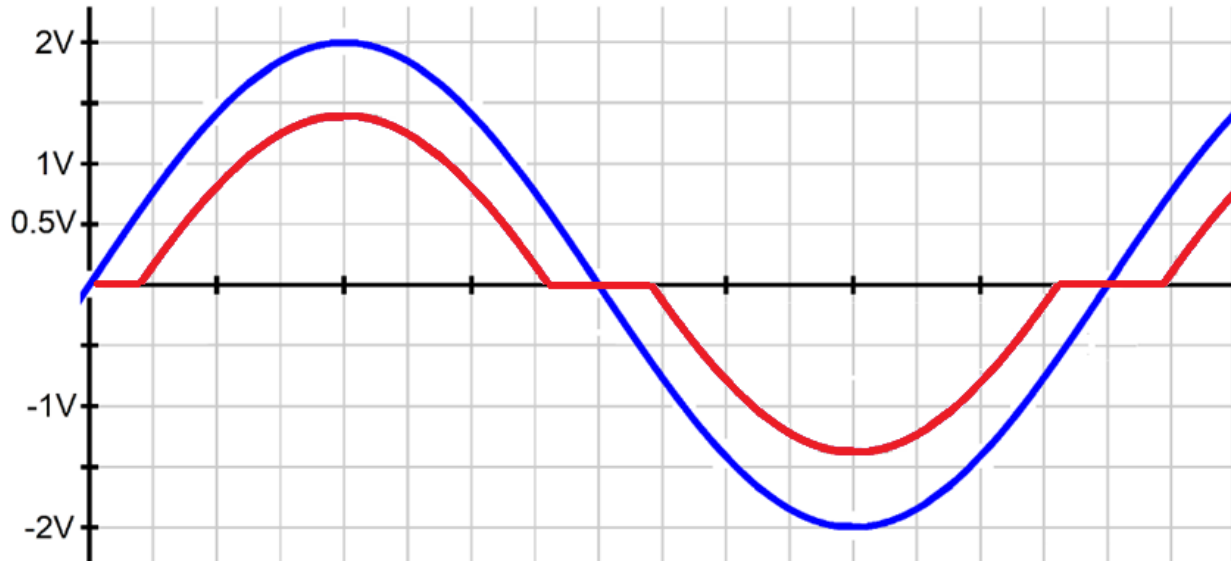
Let  $V_{in} > 0.7V$  In this case there is a path from  $V_{in}$  through the upper BJT's base emitter junction, through the  $R_L$  to ground. Thus,  $V_{out}$  is one  $V_{BE}$  (diode) drop below  $V_{in}$  or  $V_{in} - 0.7V$ . Note the lower BJT has its base emitter junction reverse biased and is off.

Let  $V_{in} = -2.0V$  In this case there is a path from GND through the load, through the lower BJT's base emitter junction, to  $V_{in}$ . Thus,  $V_{out}$  is one  $V_{BE}$  (diode) drop above -2V or  $-2V + 0.7V = -1.4V$ . Note the upper BJT has its base emitter junction reverse biased.

Let  $V_{in} < -0.7V$  In this case there is a path from GND through the load, through the lower BJT's base emitter junction, to  $V_{in}$ . Thus,  $V_{out}$  is one  $V_{BE}$  (diode) drop above  $V_{in}$  or  $V_{in} + 0.7V$ . Note the upper BJT has its base emitter junction reverse biased.

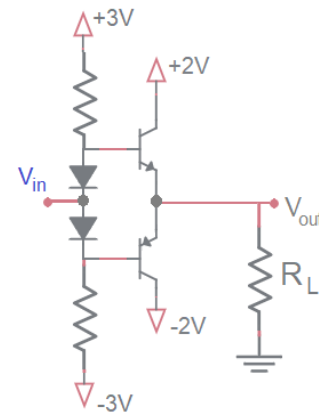
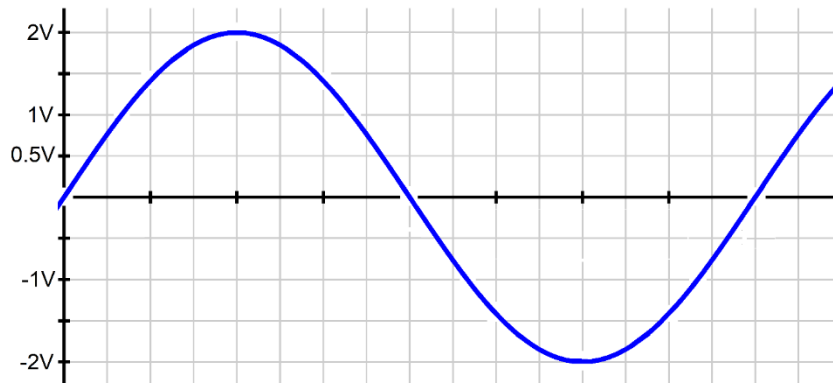
What happens when  $-0.7V > V_{in} > 0.7V$ ? Since the load is ground referenced, in the absence of any current flow,  $V_{out} = 0V$ . Since neither NPN nor PNP is forward biased in this region, there is no current flow and hence  $V_{out} = 0V$ .

This leads to the red graph below.



The flat region in the output waveform occurs when the responsibility to drive the output is handed over (crosses over) from the one BJT to the other. If this hand off has a period of time when neither BJT drives the output, then there will be a flat spot in the output, like that shown in the red waveform. This flat spot in the output is not contained in the input, hence the output is a distortion representation of the input. Hence this phenomenon is called "Cross Over Distortion".

The blue sine wave shown in the graph below is applied as the input,  $V_{in}$ , to the circuit shown below at right. Determine the output,  $V_{out}$ , and graph it on the graph below. The value of the resistors are unimportant.



The introduction of the diodes and their associated resistors makes this problem different from the previous problem and will eliminate cross over distortion at the output.

First, let's deal with the pair of resistors, one above and one below the diodes. These resistors allow current to flow from the +3V rail to the -3V rail, through the diodes. This current will establish a 0.7V drop across each diode. As  $V_{in}$  changes the voltage at the anode of the upper diode will be 0.7V above  $V_{in}$  and the voltage at the cathode of the lower diode will be 0.7V below  $V_{in}$ .

Let  $V_{in} > 0V$  In this case the anode of the upper diode is  $V_{in} + 0.7V$ . This creates a path for current to flow through the upper BJT's base emitter junction, through the  $R_L$  to ground. This means that  $V_{out}$  is one  $V_{BE}$  (diode) drop below  $V_{in} + 0.7V$  or  $V_{in}$ . Thus  $V_{out} = V_{in}$ . Note the lower BJT has its base emitter junction reverse biased and is off.

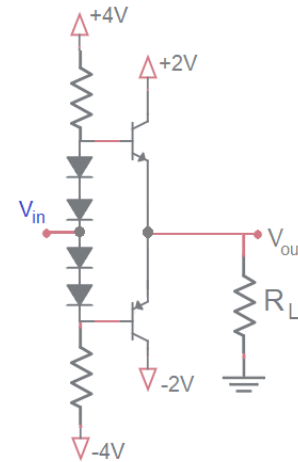
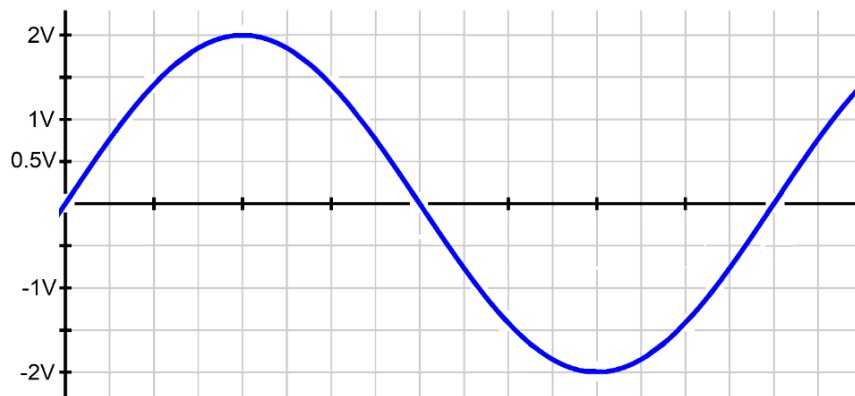
Let  $V_{in} < 0V$  In this case the cathode of the upper diode is  $V_{in} - 0.7V$ . This creates a path for current to flow from ground through the load, through the lower BJT's base emitter junction. This means that  $V_{out}$  is one  $V_{BE}$  (diode) drop above  $V_{in} - 0.7V$  or  $V_{in}$ . Thus  $V_{out} = V_{in}$ . Note the upper BJT has its base emitter junction reverse biased and is off.

Thus,  $V_{out} = V_{in}$  for  $-2V < V_{in} < 2V$

Note the +3V bias for the upper resistors is necessary because the anode of the upper diode needs to reach 2.7V in order to keep the base emitter junction of the upper BJT forward biased and the emitter at 2V. The similar argument applies to the -3V supply on the lower resistor.



The blue sine wave shown in the graph below is applied as the input,  $V_{in}$ , to the circuit shown below at right. Determine the output,  $V_{out}$ , and graph it on the graph below. The value of the resistors are unimportant.



If a little if good then more is better, right? Well the introduction of another set of diodes will introduce an unwelcome problem with this circuit.

As before the pair of resistors, one above and one below the diodes allows current to flow from the +4V rail to the -4V rail, establishing a 0.7V drop across each diode. As  $V_{in}$  changes the voltage at the anode of the upper diode (attached to the base of the NPN transistor) is 1.4V above  $V_{in}$ . The voltage at the cathode of the lower diode (attached to the base of the PNP transistor) is 1.4V below  $V_{in}$ .

Let  $-2 < V_{in} < 2V$

- The anode of the upper diode is  $V_{in} + 1.4V$ . The diode's anode is connected to the base of the upper BJT. This BJT's base emitter junction is forward biased and its emitter is at  $V_{in} + 0.7V$ . Since this BJT is ON current flows through the collector to emitter.
- The cathode of the lower diode is  $V_{in} - 1.4V$ . The diode's cathode is connected to the base of the lower BJT. This BJT's emitter base junction is forward biased and its emitter is at  $V_{in} - 0.7V$ . Since this BJT is ON current flows through the collector to emitter.

We now have a couple of problems, none of them good.

- 1) The emitters of the NPN and PNP are connected and each trying to assert different voltages. What value will actually arise? My guess is something between  $V_{in} + 0.7$  and  $V_{in} - 0.7V$ . The BJTs emitter/base diodes are going to struggle to manage the high current that are bound to arise.
- 2) Even worse? Both BJTs are ON meaning there they are conducting current through their collector/emitter junctions, forming a path from +2V to -2V. This path is low impedance and hence current will flow (short circuit style) from the +2V rail to the -2V rail bypassing the load.

It's unclear what the output voltage will be in response to the applied input, because this is an invalid circuit that will damage its components.

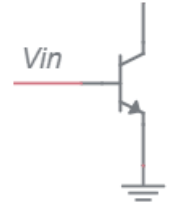


## Practical Application – Negative feedback by emitter degeneration

In the Ebers-Moll equation of a BJT's the collector current is described as

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

The  $I_S$  term, the reverse saturation current, is strongly temperature dependent. According to multiple sources, it doubles for every 10°C rise in temperature. This dependency can lead to thermal runaway in the common emitter circuit shown at right. In the following questions assume that  $V_{in}$  is constant and that the BJT is operating in the active region.



1. Look at the schematic at right, does the base emitter voltage depend on the collector current?

**Yes**

2. If the temperature of the BJT increases, does  $I_S$  increase or decrease?

**$I_S$  Increase**

3. If  $I_S$  increases, does  $I_C$  increase or decrease?

**$I_C$  Increase**

4. If  $I_C$  increases does the power dissipated by the BJT increase or decrease? Assume  $V_C$  is constant.

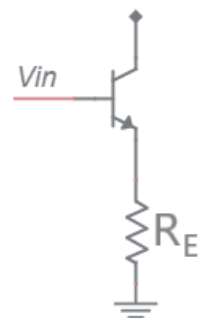
**The power dissipated increases**

5. If the power dissipated by the BJT increases, does the temperature of the BJT increase or decrease?

**The temperature increases**

From these inferences you should see that an increase in the BJT's temperature causes further increase in the temperature of the BJT – thermal runaway. This is an example of positive feedback – a tendency of a system to increase a parameter without bound. While you may like positive feedback, electrical systems can do bad things when they have too much positive feedback.

Now consider the circuit shown at right. This is a common emitter with emitter degeneration resistor. As we will see, the addition of a resistor to the emitter adds negative feedback which greatly decreases the tendency of the BJT to experience



thermal runaway. In the following questions assume that  $V_{in}$  is constant and that the BJT is operating in the active region.

1. Look at the schematic at right. Does the base emitter voltage increase or decrease when the emitter current increases?

**When the emitter current increases the base emitter voltage. Decreases.**

2. When the collector current increases, does the emitter current increase or decrease?

**When the collector current increases the emitter current increases.**

3. If the temperature of the BJT increases, does  $I_s$  increase or decrease?

**When the temperature increases  $I_s$  increases.**

4. If  $I_s$  increases, does  $I_C$  increase or decrease?

**When  $I_s$  increases then  $I_C$  increases.**

5. If  $I_C$  increases, does  $I_E$  increase or decrease?

**When  $I_C$  increases then  $I_E$  increases.**

6. If  $I_E$  increases, does  $V_E$  increase or decrease?

**When  $I_E$  increases then  $V_E$  increases.**

7. If  $V_E$  increases, does  $V_{BE}$  increase or decrease? Assume that  $V_B$  is being held constant.

**When  $V_E$  increases then  $V_{BE}$  decreases.**

8. If  $V_{BE}$  decreases, does  $I_C$  increase or decrease?

**When  $V_{BE}$  decreases then  $I_C$  decreases.**

9. If  $I_C$  decreases, does the power dissipated by the BJT increase or decrease?

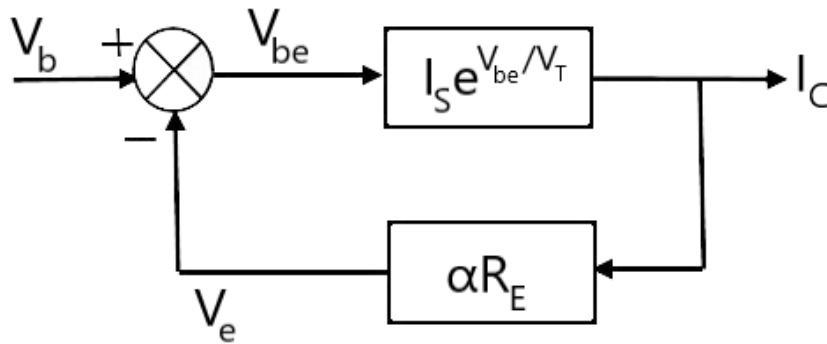
**When  $I_C$  decreases then the power dissipated by the BJT decreases.**

10. If the power dissipated by the BJT decreases, does the temperature of the BJT increase or decrease?

**When the power dissipated by the BJT decreases then the temperature of the BJT decreases.**

The ability of the decreasing  $V_{BE}$  being able to keep the increasing  $I_S$  in check depends on the magnitude of the emitter resistor. If the emitter resistor is too small, then the BJT may still exhibit thermal runaway. In that case, you should increase the emitter resistor until thermal runaway is kept in check.

Now, let's see if you can make a feedback diagram to describe this negative feedback using the diagram below. Annotate the diagram according to the instructions below.



1. Make the reference input equal to  $V_b$  (this is the input at the left).
2. What must the inverting input to the summer junction be for the output of the summer junction to be  $V_{be}$ ? Write this signal and  $V_{be}$  on the diagram.
3. Fill in the upper block with the right-hand side of the Ebers-Moll equation.
4. Label the output of the block diagram with the left-hand side of the Ebers-Moll equation. This should make sense because the Ebers-Moll equation is a bunch of constants with two variables.
5. Fill in the lower block with an equation which converts the block diagram output into the variable used in the inverting input to the summer junction. Hint, its Ohm's law.

This block diagram shows the  $V_{BE}$  variable on the forward path and the  $R_E$  variable on the backward path. The interplay between these two determines the thermal stability of the common emitter with emitter degeneration resistor circuit.

As a general rule of thumb, to stabilize a common emitter use a resistor value that increases by 0.3V to 0.5V at the maximum current you expect through the BJT.

<To Be Completed>

### Practical Application – Thermal Runaway in parallel BJT circuits.

You've been assigned to illuminate a Cree Xlamp XM-L2 High Power LED with 1 Amp, but all you have at hand are some 2N3904 NPN BJTs in a TO-92 package – the same kind we use in the lab. You look through the datasheet for the 2N3904 and find the maximum ratings table shown at right. Since you know that LEDs are current controlled, you focus on the maximum collector current row.

MAXIMUM RATINGS

| Rating                                                                                    | Symbol         | Value       | Unit                       |
|-------------------------------------------------------------------------------------------|----------------|-------------|----------------------------|
| Collector – Emitter Voltage                                                               | $V_{CE0}$      | 40          | Vdc                        |
| Collector – Base Voltage                                                                  | $V_{CBO}$      | 60          | Vdc                        |
| Emitter – Base Voltage                                                                    | $V_{EBO}$      | 6.0         | Vdc                        |
| Collector Current – Continuous                                                            | $I_C$          | 200         | mA <sub>dc</sub>           |
| Total Device Dissipation<br>@ $T_A = 25^\circ\text{C}$<br>Derate above $25^\circ\text{C}$ | $P_D$          | 625<br>5.0  | mW<br>mW/ $^\circ\text{C}$ |
| Total Device Dissipation<br>@ $T_C = 25^\circ\text{C}$<br>Derate above $25^\circ\text{C}$ | $P_D$          | 1.5<br>12   | W<br>mW/ $^\circ\text{C}$  |
| Operating and Storage Junction Temperature Range                                          | $T_J, T_{stg}$ | -55 to +150 | $^\circ\text{C}$           |

What is the maximum continuous current that you can run through a 2N3904?

In order to get the required 1 Amp, you decide to parallel 5 2N3904's in parallel.